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BEST PRACTICE SERIES

Goals & Objectives for Preventive Maintenance

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Global Mining

April 07
1006-3.1-1022

1.0 Introduction

Site support processes such as Preventive Maintenance are often times very much task-oriented, that is, the individuals responsible for their execution are aware of the activities associated with the particular job but are unaware of the contributions their area of responsibility is making to the overall success of the project. Using Performance Metrics to help define the specific impact that their role plays in the success of the project tends to make employees more goal-oriented and less task-oriented significantly increasing their personal involvement, sense of ownership and consequently performance in their jobs.

This Best Practice relates to **version II of Site Assessment items 2.9.1 and 2.9.8.**

2.0 Best Practice Description

A condition-based maintenance program is comprised of a number of critical functions. Due to the frequency with which it is performed and the overall impact it can have on the success of a mine site, one of the most important of those functions is Preventive Maintenance (PM). Unfortunately, the term Preventive Maintenance has become almost generic in that it is used to describe virtually any maintenance, service or repair activity not performed as a result of a breakdown or failure.

Preventive Maintenance commonly involves a number of activities completed within the same time frame or window of opportunity. These activities normally include, but are not limited to the following:

- Machine washing (either complete or partial washing),
- Lubrication service (changing oil and filters and greasing of required locations),
- Fluids management (taking oil samples, inspecting magnetic plugs, screens or filters),
- Inspections (visual) for defects, damage, and wear,
- Conducting instrumented system tests or measurements of performance, and
- Execution of repairs for known defects and problems.

When the term Preventive Maintenance is used, care must be taken to establish and understand which of the above activities are actually included.

Machine downtime is a function of frequency and duration shutdown events. Because PM is a high frequency activity (typically done every 250 to 500 hours), it is a major contributor to machine downtime particularly when PM duration (affected by efficiency) is long. This explains why PM efficiency should be a primary concern. Another equally important concern should be PM quality / effectiveness ... getting the maximum possible benefit from the PM downtime event.

Unfortunately, many mines and, as a consequence, the individuals responsible for performing Preventive Maintenance at those mines, view it as repetitive, routine, boring and, in many instances, almost punitive duty. When this is the case, PM execution in terms of efficiency and more often effectiveness suffers and the value of Preventive Maintenance is greatly diminished.

One way to re-engage PM personnel is to assist them in their understanding and appreciation of the importance of Preventive Maintenance. Defining Key Performance Indicators (KPI's) that relate to the efficiency and effectiveness of the PM function and training the staff on the impact of those KPI's on overall site operations and success will encourage the staff to become more accountable for their performance and should drive improvement in the PM process.

KPI's that quantify and contribute to an understanding of PM efficiency and effectiveness are:

- Mean Time Between Shutdowns after PM (MTBS $_{\text{after PM}}$): A measure of PM effectiveness and a clear indication of the quality of the job performed. Establish a target for hours to the first shutdown after PM and report back to the PM crew not only their actual performance relative to the target but also the cause of the first stoppage so issues that are found to be chronic can be addressed through changes in the PM process. When armed with this kind of information the PM staff should be capable of providing better (more reliable) performance in terms of MTBS after the equipment is returned to operation. The Benchmark (best in class) for large Off Highway Trucks is 105 hours; a realistic target is 2 to 3 times the overall MTBS.

- Unavailability due to PM: A measure that quantifies the impact of PM on availability. There is no Benchmark associated with this metric but a reasonably valid target is in the range of 2.75 to 3.25% for large Off Highway Trucks in the 785 – 793 size class. The result of this measure should be taken in context with MTBS $_{\text{after PM}}$. If unavailability due to PM is below the range and MTBS $_{\text{after PM}}$ is low, it is a safe assumption that insufficient time and effort is being placed on PM. Conversely, if unavailability due to PM is above the range and MTBS $_{\text{after PM}}$ is high, one can assume that the site is placing substantial emphasis on the value of PM.

- Mean Time To Repair for PM (MTTR $_{\text{PM}}$): Average PM execution time; a measure of PM efficiency. There is no Benchmark associated with this measure but a reasonable target is in the 7.75 to 8.5 hour range for large Off Highway Trucks that are on a 250 hour PM service interval. The target for equipment that is on a 500 hour PM service interval is higher, i.e. double that of the 250 hour interval. Establish target times for PM completion times and compare actual PM execution times relative to those targets.

- Service Accuracy (SA): A measurement of Preventive Maintenance execution timeliness based on a statistical calculation that predicts the probability that the next PM service will occur within the recommended range (+/- 25 hours of target interval). The calculation is based upon past performance and assumes that PM intervals are normally distributed about the mean. The Benchmark for S.A. is 95% but an aggressive target that will yield excellent results is 90%.

- Backlogs executed during PM: Quantifies the use of the "window of opportunity" presented during the PM shutdown.

- Backlogs generated during PM: Quantifies the use of the "window of opportunity" presented during the PM shutdown for defect detection (an element of Condition Monitoring).

3.0 Implementation Steps

Critical steps associated with the development and implementation of a performance-based Preventive Maintenance process include:

- PM Strategy:
  - ✓ Define what metrics will be used to quantify PM performance.
  - ✓ Establish realistic performance targets for those metrics.
  - ✓ Determine the raw data required to make the necessary calculations for quantifying the metrics identified above (turn the data into information you can use to help manage the site).
  - ✓ Define the distribution network necessary to communicate the information.
  - ✓ Define PM report format, display, and frequency requirements.

- Personnel:
  - ✓ Determine what individuals in the organization will be responsible to the acquisition of raw data, the calculation of the required performance measures, and the distribution of that information. Much of this activity will take place in the Planning Department at most sites; the planner, “Fleet Analyst” (reliability engineer) are key individuals involved in on-site data management, i.e. calculations, analysis / interpretation, and distribution.
  - ✓ The scheduler must be aware of results interpretation guidelines and have access to the results since the nature of his work puts him in a position to drive positive change.
  - ✓ Define the individuals who will be responsible for PM. Include not only Technicians but also their Management and Supervision. (It is our belief that PM is important enough to justify developing “specialists” for the PM activity).
  - ✓ Determine who will be responsible for training the organization in the analysis and interpretation of PM performance results; include site management and everyone involved in the execution of Preventive Maintenance activities.
  - ✓ Determine what individual(s) from the PM organization will represent the interests of PM in Continuous Improvement process.

- Implementation:
  - ✓ Deliver the training on the analysis and interpretation of PM performance results.
  - ✓ Establish a specific location in the area that Preventive Maintenance is executed for the display of PM performance results. Dedicated bay(s) assigned to PM activities is preferred.
  - ✓ Publish and display PM performance results in the designated area.
  - ✓ Use the output from the analysis and interpretation of PM performance results as input to the on-site Continuous Improvement process … devise solutions to identified problems.
  - ✓ Take appropriate action(s) to implement solutions identified in the Continuous Improvement process.

Photo 1: PM performance and scheduling board in the PM bay at a mine in Brazil.

Photo 2: A PM Technician explaining his interpretation of PM performance.

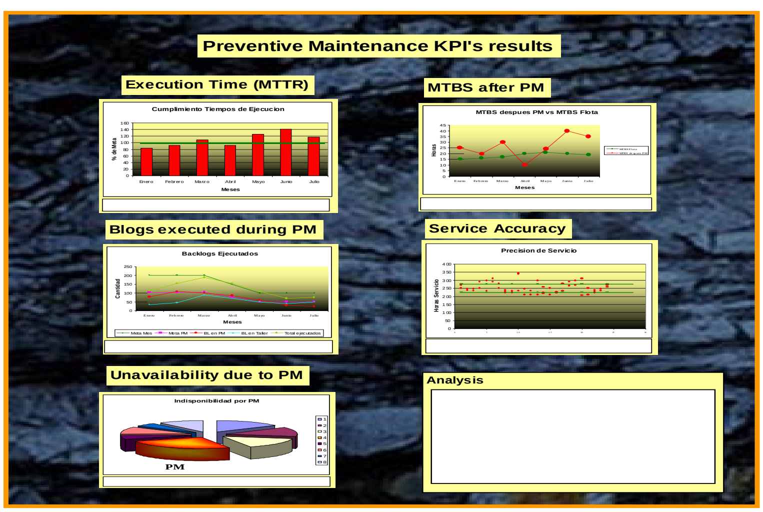

Photo 3: Schematic of the section of the PM Board dedicated to display the PM KPI's. PM performance results.

4.0 Benefits

Benefits are accrued as a result of the influence of increased PM performance on fleet reliability, that is improved overall MTBS, and are linked to the impact of increased PM effectiveness and efficiency and the resultant overall improvement of fleet availability.

The payback period is based on the fleet size, initial (pre-implementation) level of PM performance, and the extent of improvement realized and can be quantified in terms of increased machine availability (additional productive hours), improved resource deployment (personnel and facilities), and the reduction or elimination of penalties related to performance guarantee shortfalls.

5.0 Resources Required

- Human Resources: Human resource requirements can be identified through a review of the specific steps outlined in the "Personnel" section of the "Implementation Steps" guidelines above. The majority of those resource requirements will be contained within the Preventive Maintenance organization of the Maintenance Department, the Maintenance Planning organization and the Training department.
- Facility Resources: A bulletin board or some other medium to display and communicate results of the Preventive Maintenance process will be required to implement this Best Practice.

Costs include the physical implementation of a bulletin board to display and communicate results of the Preventive Maintenance process, the time spent developing materials and training the PM staff in the analysis and interpretation of those results, and the implementation of related changes developed via the Continuous Improvement process.

6.0 Supporting Attachments

Metrics (KPI's) to Assess Process Performance –PDF document (see Attachments tab)

7.0 Related Best Practices

None

8.0 Acknowledgements

This *Goals & Objectives for Preventive Maintenance* Best Practice was authored by:

Jim McCaherty
Mining Equipment Management / Processes Division
Caterpillar Global Mining
+1 309 675 5595
McCaherty_James_W@cat.com

We would also like to acknowledge Sotreq's Mining Division for their willingness to share their accomplishments with Caterpillar mining dealers worldwide.

Special recognition to:

Pedro Perim – Sotreq – Mining Business & Operations Support Manager
Alexandre Renno - Sotreq - Mining Project Support Mgr (formerly Sotreq Project Mgr at Samarco)